Design, Fabrication, and Data Analysis of Mach-Zehnder Interferometer

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In this report, Mach–Zehnder Interferometers (MZIs) are designed and fabricated with a variety of imbalance lengths. The effect of imbalance length on the spectral response of the MZI is investigated, and fabrication imperfections introduced during the fabrication process are also analyzed. Key performance parameters, including effective index, group index, and free spectral range (FSR), are compared between simulation and experimental results. This comparison provides insight into the accuracy of the design, the impact of fabrication tolerance, and the overall performance of the fabricated MZI devices.

1. INTRODUCTION

MZIs are one of the most fundamental and widely used building blocks in integrated photonics. Due to their simple structure, high design flexibility, and strong phase sensitivity, MZIs have been extensively employed in optical modulation, filtering, switching, sensing, and wavelength-division multiplexing. In silicon photonic integrated circuits (PICs), MZIs are particularly attractive because they can be fabricated using CMOS-compatible processes, enabling compact, scalable, and low-cost optical devices for communication, sensing, and signal-processing applications.

A conventional MZI consists of an input optical splitter, two interferometer arms, and an output optical combiner. The input optical signal is divided into two paths, where each arm accumulates an optical phase determined by the effective refractive index and physical length of the waveguide. When the two optical fields recombine at the output, the resulting transmission depends on the relative phase difference between the two arms. This phase difference can be expressed as,

$$\Delta\phi = \frac{2\pi}{\lambda} n_{\text{eff}} \Delta L. \tag{1}$$

where λ is the operating wavelength, n_{eff} is the effective refractive index of the guided mode, and ΔL is the length difference between the two arms. Depending on the path-length difference, MZIs can be classified as balanced or imbalanced. In a balanced MZI, the two interferometer arms have equal lengths, meaning that $\Delta L = 0$. Ideally, both optical signals experience the same propagation distance and accumulate the same phase. As a result, the output transmission is relatively insensitive to wavelength-dependent phase variation caused by path-length difference. Balanced MZIs are commonly used in applications such as optical switching, modulation, and phase-sensitive measurements, where external phase tuning or active control is often required to produce constructive or destructive interference. In contrast, an imbalanced MZI has unequal arm lengths, where $\Delta L \neq 0$. The additional optical path length in one arm introduces a wavelength-dependent phase difference between the two arms. As a result, the output transmission exhibits periodic constructive and destructive interference fringes as a function of wavelength. Compared with a balanced MZI, an imbalanced MZI offers several advantages for passive photonic characterization and wavelength-dependent analysis. Since the imbalance length naturally produces periodic spectral fringes, the device can be used to extract important waveguide parameters such as FSR, group index, effective index, and dispersion from the measured transmission spectrum. Furthermore, by designing MZIs with different imbalance lengths, the dependence of spectral response on path-length difference can be systematically studied. Larger imbalance lengths produce smaller FSR values and more closely spaced fringes, while shorter imbalance lengths produce wider spectral spacing. This makes imbalanced MZIs particularly useful as test structures for evaluating waveguide properties, validating simulation models, and analyzing fabrication-induced variations.

In this report, MZIs with different arm lengths are designed, fabricated, and experimentally characterized. The transmission spectra of the fabricated devices are analyzed to extract key performance parameters, including effective index, group index, and FSR. These experimental results are then compared with simulation data to evaluate the agreement between design and fabrication.

2. THEORETICAL BACKGROUND OF THE MZI PIC

A MZI PIC is an interferometric structure that converts an optical phase difference into a measurable intensity variation. As shown in Fig. 1(a), the PIC MZI consists of several passive optical components, including input and output grating couplers, optical Y splitter, waveguide arms, and directional coupler, as shown in Figs. 1(b)-(d). Light from a tunable laser is first coupled into the chip through an input grating coupler. The optical signal is then divided into two waveguide arms, propagates through different optical paths, and is recombined at the output. Depending on the relative phase accumulated in the two arms, the recombined fields interfere either constructively or destructively, producing a wavelength-dependent transmission spectrum.

The input and output grating couplers are used to couple light between an external optical fiber or free-space optical beam and the on-chip silicon waveguide. Since the mode size of an optical fiber is much larger than that of a silicon waveguide, direct coupling is inefficient. A grating coupler overcomes this mismatch by using a periodic diffractive structure to scatter light into or out of the waveguide under a phase-matching condition. In the measurement configuration, the input grating coupler receives light from the tunable laser, while the output grating couplers direct the transmitted optical signals toward the detectors. Therefore, the grating couplers strongly influence the total insertion loss, operating bandwidth, and polarization sensitivity of the measured MZI response.

After light is coupled into the waveguide, an optical power splitter divides the input field into two arms. For an ideal 50:50 splitter, the input optical intensity is equally distributed between the two paths,

$$I_1 = I_2 = \frac{I_{\text{in}}}{2}. \tag{2}$$

Since optical intensity is proportional to the square of the electric field amplitude, the corresponding electric fields in the two arms are,

$$E_1 = E_2 = \frac{E_{\text{in}}}{\sqrt{2}}. \tag{3}$$

The waveguide arms form the phase-accumulation region of the MZI. In silicon photonics, the waveguide typically consists of a high-index silicon core surrounded by a lower-index silicon dioxide cladding. This high refractive-index contrast provides strong optical confinement and allows compact routing of light on the chip. The optical field propagating along a waveguide can be written as,

$$E(z,t) = E_0 e^{j(\omega t - \beta z)}, \tag{4}$$

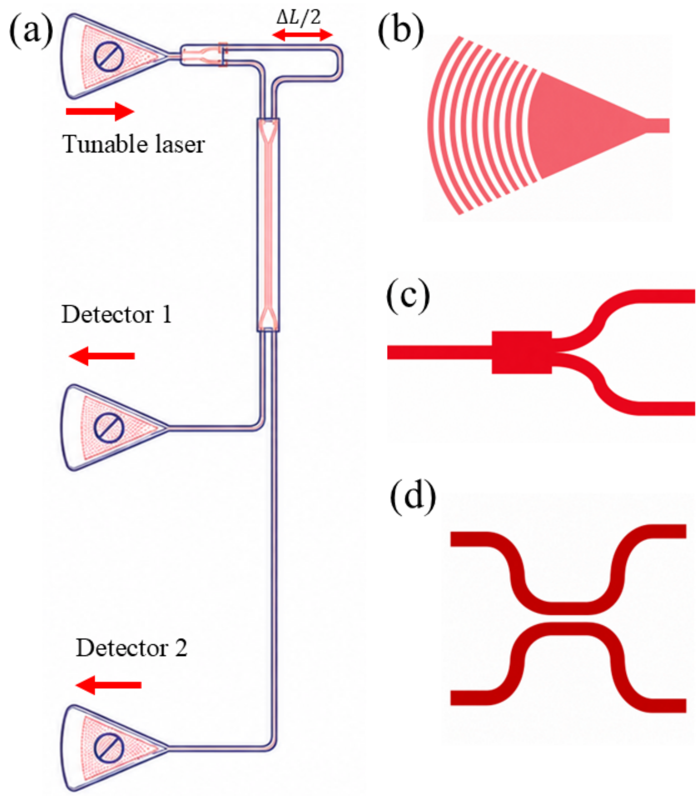

Fig. 1. Schematic illustration of the photonic integrated MZI and its main building blocks. (a) Complete MZI layout, where light from a tunable laser is coupled into the chip via grating coupler, split into two interferometer arms through splitter with an imbalance length of ΔL, recombined, and collected at the output detectors via grating couplers. (b) Grating coupler used for optical coupling between the external fiber beam and the on-chip waveguide. (c) Optical splitter used to divide the input optical power into two waveguide arms. (d) Directional coupler used for optical coupling and recombination of the propagating fields.

Here, E_0 is the electric field amplitude, ω is the angular frequency, β is the propagation constant, z is the propagation direction, and t is time. The propagation constant of a guided mode is given by,

$$\beta = \frac{2\pi n_{\text{eff}}}{\lambda}, \tag{5}$$

Where, n_{eff} is the effective refractive index of the guided mode, and λ is the operating wavelength. Unlike the bulk refractive index, the effective index depends not only on material properties but also on waveguide geometry, polarization, wavelength, and modal confinement. The phase accumulated by light propagating through a waveguide of length L is,

$$\phi = \beta L. \tag{6}$$

Therefore, for an MZI with two arms of lengths L_1 and L_2, the path-length difference is,

$$\Delta L = L_1 - L_2, \tag{7}$$

The corresponding phase difference between the two arms is,

$$\Delta\phi = \beta \Delta L = \frac{2\pi n_{\text{eff}} \Delta L}{\lambda}. \tag{8}$$

In an imbalanced MZI, the two arms have unequal lengths, so $\Delta L \neq 0$. The additional path length introduces a wavelength-dependent phase difference between the two arms. Consequently, the output transmission exhibits periodic interference fringes as a function of wavelength. For an ideal lossless imbalanced MZI with a 50:50 splitter and combiner, the normalized output intensity can be expressed as,

$$\frac{I_{\text{out}}}{I_{\text{in}}} = \frac{1}{2}\left[1 + \cos(\beta \Delta L)\right]. \tag{9}$$

Constructive interference occurs when

$$\Delta\phi = 2m\pi, \tag{10}$$

while destructive interference occurs when

$$\Delta\phi = (2m+1)\pi, \tag{11}$$

Where, m is an integer.

The output combiner converts the phase difference between the two arms into an intensity variation. If the two optical fields arrive in phase, constructive interference produces maximum transmission. If they arrive out of phase, destructive interference produces minimum transmission. The recombined electric field at one output port can be written as,

$$E_{\text{out}} = \frac{1}{\sqrt{2}}\left(E_1 + E_2\right), \tag{12}$$

The measured output intensity is proportional to

$$I_{\text{out}} \propto |E_{\text{out}}|^2. \tag{13}$$

The output power is commonly expressed in decibels as,

$$T_{\text{dB}} = 10 \log_{10}\left(\frac{I_{\text{out}}}{I_{\text{in}}}\right). \tag{14}$$

The periodicity of the transmission spectrum of an imbalanced MZI is described by the free spectral range (FSR), which is the wavelength spacing between two adjacent transmission maxima or minima,

$$\text{FSR} = \lambda_{m+1} - \lambda_m. \tag{15}$$

The FSR is related to the group index of the waveguide and the imbalance length by,

$$\text{FSR} = \frac{\lambda^2}{n_g \Delta L}, \tag{16}$$

Here, n_g is the group index of the guided mode. The group index is connected to the wavelength dependence of the effective index through,

$$n_g = n_{\text{eff}} - \lambda \frac{dn_{\text{eff}}}{d\lambda}. \tag{17}$$

Therefore, the group index can be extracted from the measured FSR using by,

$$n_g = \frac{\lambda^2}{\text{FSR} \cdot \Delta L}. \tag{18}$$

3. RESULTS AND DISCUSSIONS

Lumerical MODE was used to simulate the guided modes of the silicon waveguide and to extract the wavelength-dependent effective index and group index. As shown in Fig. 2(a), the effective index decreases as the wavelength increases from 1500 to 1600 nm. This behavior occurs because the optical mode becomes less tightly confined inside the high-index silicon core at longer wavelengths, causing a larger fraction of the optical field to extend into the lower-index cladding region. In contrast, the group index shows a slight increasing trend over the same wavelength range. The group index is related to the effective index dispersion and can be expressed as,

$$n_g = n_{\text{eff}} - \lambda \frac{dn_{\text{eff}}}{d\lambda}. \tag{19}$$

Therefore, both the absolute value of n_{eff} and its wavelength dependence contribute to the extracted group index.

Figures 2(b) and (c) show the simulated optical mode profiles of the guided modes. The fundamental quasi-TE mode exhibits a higher effective index of $n_{\text{eff}} = 2.442$ and a smaller effective mode area of $A_{\text{eff}} = 0.192\ \mu\text{m}^2$, indicating stronger confinement inside the silicon waveguide. The waveguide confinement factor for this mode is $\Gamma_{\text{wg}} = 0.78$, meaning that a large portion of the optical field is confined within the waveguide core. In comparison, the quasi-TM mode has a lower effective index of $n_{\text{eff}} = 1.771$, a larger effective mode area of $A_{\text{eff}} = 0.344\ \mu\text{m}^2$, and a smaller confinement factor of $\Gamma_{\text{wg}} = 0.44$. This indicates weaker confinement and stronger field penetration into the surrounding cladding.

The extracted modal parameters are important for the design and analysis of the MZI. In particular, the effective index determines the phase accumulated by light propagating through each interferometer arm, while the group index is directly related to the free spectral range of the imbalanced MZI. Thus, the Lumerical MODE simulation provides the fundamental waveguide parameters required for predicting the spectral response of the PIC MZI.

The final photonic layout was designed and inspected using KLayout before fabrication. Figure 3(a) shows the complete class layout, which contains multiple photonic devices, test structures, calibration structures, and alignment features arranged within the available fabrication area. The selected region, indicated by the dashed red box, corresponds to the area where my designed MZI structures are placed. This full-chip layout ensures that different photonic components can be fabricated simultaneously and characterized under the same process conditions.

Figure 3(b) shows the enlarged view of the designed MZI layout. Four imbalanced MZI structures were included with different path-length differences, ΔL, in order to study the effect of imbalance length on the spectral response. Each MZI consists of an input grating coupler, an optical splitter, two waveguide arms with a designed length difference, and output grating couplers connected to the measurement ports. The imbalance length introduces a wavelength-dependent phase difference between the two arms, which produces periodic interference fringes in the transmission spectrum. According to the MZI FSR relation, a larger imbalance length results in a smaller FSR, while a shorter imbalance length produces a larger FSR. Therefore, the designed MZIs with $\Delta L = 81\ \mu$m, 100 μm, and 150 μm allow the dependence of FSR, group index, and fabrication tolerance on path-length imbalance to be systematically analyzed. The repeated $\Delta L = 81\ \mu$m MZI design can also be used to check device repeatability and fabrication uniformity. If two nominally identical MZIs show different spectral responses after measurement, the difference may arise from fabrication imperfections such as waveguide width variation, sidewall roughness, etch-depth variation, or grating-coupler mismatch. Thus, the designed layout provides both wavelength-dependent MZI characterization and a practical way to evaluate fabrication-induced variations in the PIC platform.

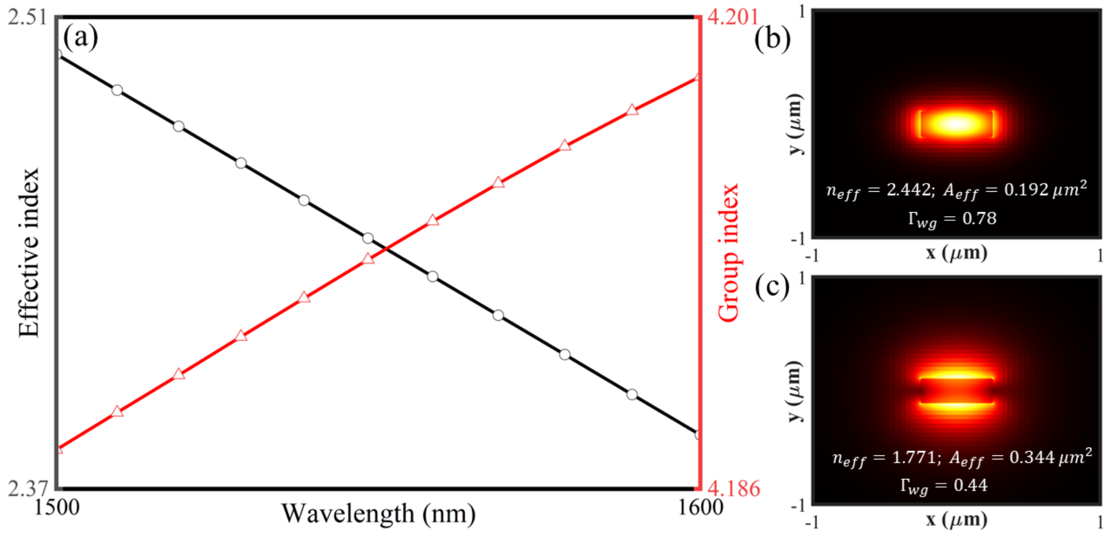

Fig. 2. Lumerical MODE simulation results for the silicon strip waveguide. (a) Wavelength-dependent effective index and group index extracted from the guided mode over the wavelength range of 1500 - 1600 nm. (b) Simulated optical mode profile of the fundamental quasi-TE mode with $n_{\text{eff}} = 2.442$, $A_{\text{eff}} = 0.192\ \mu\text{m}^2$, and $\Gamma_{\text{wg}} = 0.78$. (c) Simulated optical mode profile of the quasi-TM mode with $n_{\text{eff}} = 1.771$, $A_{\text{eff}} = 0.344\ \mu\text{m}^2$, and $\Gamma_{\text{wg}} = 0.44$.

Figure 4 shows the transmission spectra of the fabricated imbalanced MZI over the wavelength range of 1500 - 1580 nm. The periodic maxima and minima in the spectrum originate from the wavelength-dependent phase difference between the two interferometer arms. When the two optical fields recombine in phase, constructive interference occurs and the transmission becomes maximum. In contrast, when the phase difference satisfies the destructive interference condition, sharp transmission minima are observed. In Fig. 4(a), the measured transmission shows deep interference dips, indicating strong destructive interference between the two arms of the MZI. The processed curve follows the main spectral trend and helps identify the interference features more clearly. Figure 4(b) shows the measured and smoothed/fitted transmission response, where the periodic behavior is preserved after processing. Small deviations between the measured and fitted curves may arise from fabrication imperfections, grating-coupler loss, waveguide sidewall roughness, splitter imbalance, and measurement noise. Overall, the observed periodic transmission response confirms the successful operation of the fabricated imbalanced MZI and enables extraction of important parameters such as FSR and group index.

Figure 5 compares the experimental and simulated transmission spectra of the designed imbalanced MZIs with different imbalance lengths. The experimental spectra are shown in Figs. 5(a) and (c), while the corresponding simulated spectra are shown in Figs. 5(b) and (d). For all devices, periodic transmission maxima and minima are observed over the wavelength range of 1500 - 1580 nm. These spectral fringes are produced by the wavelength-dependent phase difference between the two arms of the MZI. As the wavelength changes, the phase difference also changes, resulting in alternating constructive and destructive interference at the output. Therefore, increasing ΔL results in a smaller FSR and more closely spaced interference fringes, while decreasing ΔL produces a larger FSR.

The simulated spectra in Figs. 5(b) and (d) clearly show the expected dependence of fringe spacing on imbalance length. The MZI with the largest imbalance length exhibits the most closely spaced fringes, whereas the device with the shortest imbalance length shows wider spectral spacing. The experimental spectra show the same general trend, confirming that the fabricated MZIs operate as

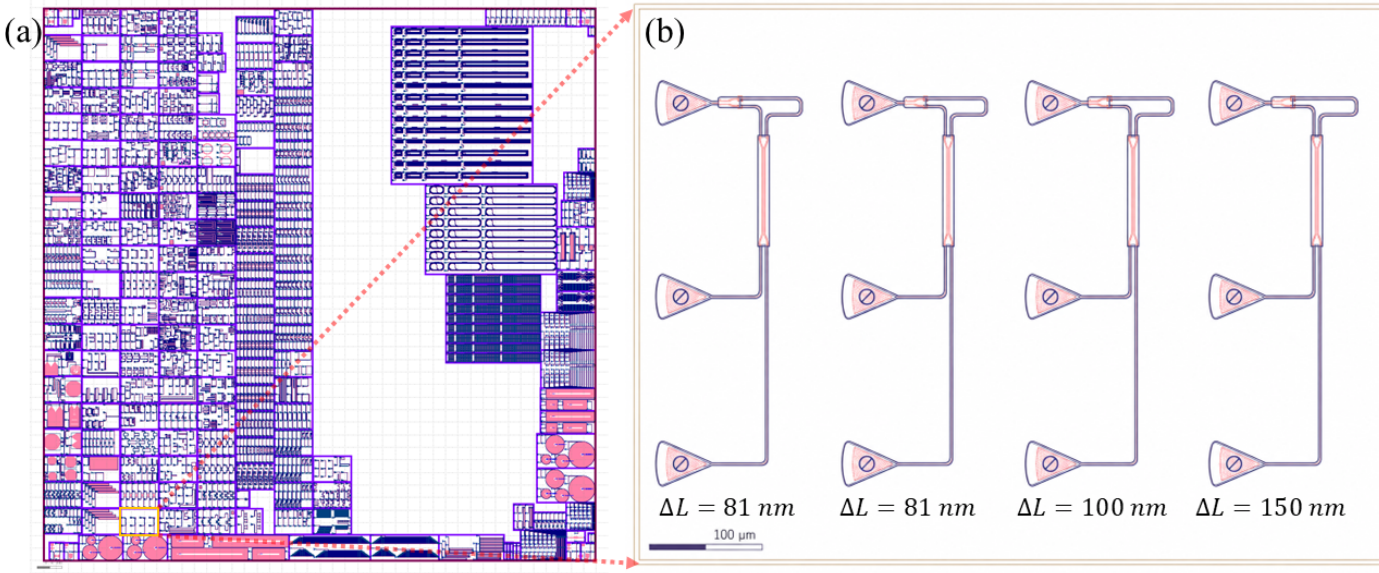

Fig. 3. KLayout layout of the photonic integrated circuit and the designed MZIs. (a) Complete class layout containing different photonic test structures and design blocks used for fabrication. (b) Enlarged view of the designed MZI section, consisting of four imbalanced MZI devices with different path-length differences of $\Delta L = 81$ μm, 81 μm, 100 μm, and 150 μm. Each MZI includes input/output grating couplers, optical splitters, waveguide arms, and output ports for spectral characterization.

expected. However, some differences are observed between the experimental and simulated results. The experimental spectra contain a wavelength-dependent transmission envelope, nonuniform dip depths, and small shifts in the interference fringes. These deviations can be attributed to fabrication imperfections, waveguide width variation, etch-depth variation, sidewall roughness, grating-coupler loss, splitter imbalance, bending loss, and measurement noise. In contrast, the simulated spectra are more regular because they are generated using ideal or nominal design parameters. Overall, the agreement between the experimental and simulated spectra confirms the functionality of the designed imbalanced MZIs and enables extraction of important waveguide parameters such as FSR and group index.

Figure 6 presents the extracted effective index, group index, and FSR of the imbalanced MZIs and compares the simulation results with the experimental measurements. The optical parameters were extracted over the wavelength range of 1500 - 1580 nm to evaluate the wavelength-dependent behavior of the fabricated MZI devices. As shown in Fig. 6(a), the effective index decreases with increasing wavelength for all MZI designs. This decreasing trend is mainly caused by waveguide dispersion, where the guided optical mode becomes less confined inside the silicon core at longer wavelengths. The effective index determines the phase accumulated by light as it propagates through each arm of the MZI and is given through the propagation constant,

$$\beta = \frac{2\pi n_{\text{eff}}}{\lambda}, \tag{20}$$

Where, β is the propagation constant, n_{eff} is the effective index, and λ is the wavelength. Figure 6(b) shows the extracted group index as a function of wavelength. The group index is related to the wavelength dependence of the effective index and can be expressed as,

$$n_g = n_{\text{eff}} - \lambda \frac{dn_{\text{eff}}}{d\lambda}. \tag{21}$$

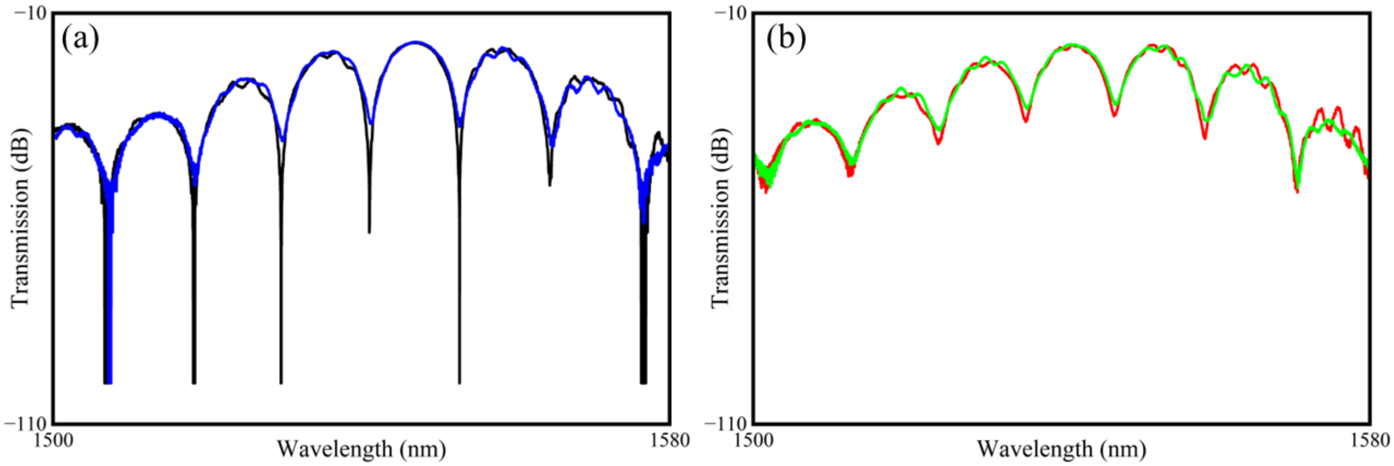

Fig. 4. Transmission spectra of the fabricated imbalanced MZI measured over the wavelength range of 1500 - 1580 nm. (a) Measured transmission response showing periodic interference minima, where the black curve represents the measured spectrum and the blue curve represents the processed or fitted response. (b) Comparison between the measured transmission spectrum and the smoothed/fitted response after spectral processing. The periodic dips correspond to destructive interference between the two MZI arms.

The group index is an important parameter for analyzing the spectral response of an imbalanced MZI because it directly determines the FSR. The simulated and experimental group-index values show similar trends, although small deviations are observed. These differences may originate from fabrication imperfections such as waveguide width variation, etch-depth variation, sidewall roughness, and uncertainty in the measured fringe positions. Figure 6(c) shows the FSR extracted from the transmission spectra. The FSR is defined as the wavelength spacing between two adjacent interference peaks or dips,

$$\mathrm{FSR} = \lambda_{m+1} - \lambda_m. \tag{22}$$

For an imbalanced MZI, the FSR is related to the group index and imbalance length by,

$$\mathrm{FSR} = \frac{\lambda^2}{n_g \Delta L}, \tag{23}$$

Here, ΔL is the path-length difference between the two interferometer arms. This relation shows that the FSR is inversely proportional to the imbalance length. Therefore, the MZI with the shortest imbalance length exhibits the largest FSR, while the MZI with the longest imbalance length exhibits the smallest FSR. Overall, the simulation and experimental results show good qualitative agreement. The small mismatch between the two results can be attributed to fabrication tolerances, grating-coupler response, splitter imbalance, waveguide loss, and measurement noise. These comparisons confirm that the fabricated imbalanced MZIs can be used to extract important waveguide parameters and to evaluate the accuracy of the design and fabrication process.

4. CONCLUSION

In this report, MZIs with different imbalance lengths were designed, simulated, fabricated, and experimentally analyzed. The theoretical operating principle of the MZI was first discussed, where the wavelength-dependent phase difference between two interferometer arms produces periodic

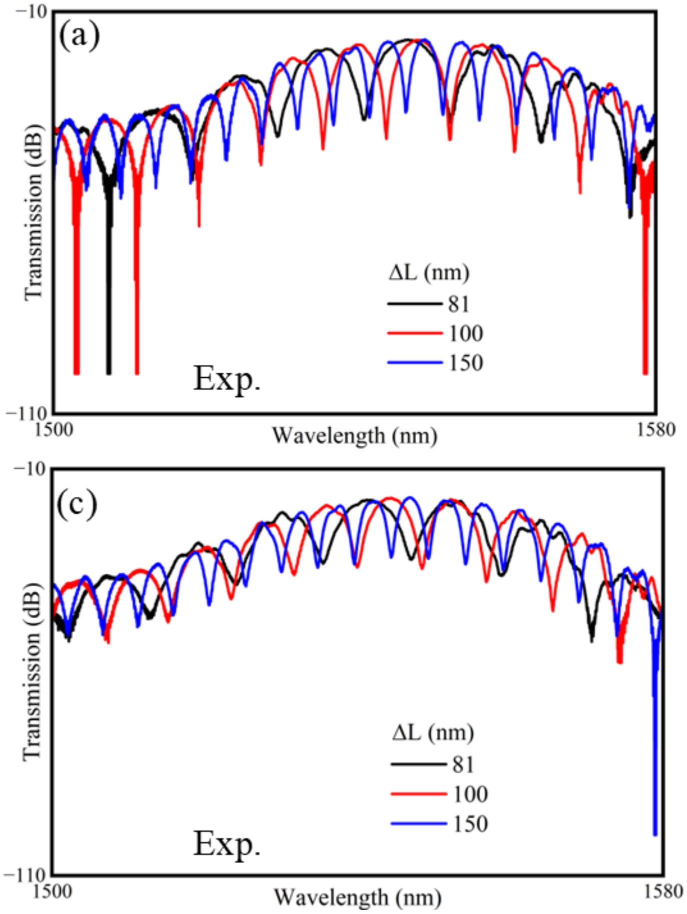

Fig. 5. Comparison of experimental and simulated transmission spectra of imbalanced MZIs with different imbalance lengths. (a) Experimental transmission spectra for MZIs with $\Delta L = 81$, 100, and 150. (b) Simulated transmission spectra for the corresponding imbalance lengths. (c) Experimental spectra after spectral processing/normalization for comparison of fringe positions and periodicity. (d) Simulated spectra showing the expected wavelength-dependent interference response. The black, red, and blue curves correspond to $\Delta L = 81$, 100, and 150, respectively.

constructive and destructive interference. The main building blocks of the PIC MZI, including grating couplers, optical splitters, waveguide arms, and directional couplers, were also described.

Lumerical MODE was used to extract the fundamental waveguide parameters, including the effective index, group index, mode profile, effective mode area, and waveguide confinement factor. The simulation results showed that the effective index decreases with increasing wavelength, while the group index exhibits wavelength-dependent behavior due to waveguide dispersion. These simulated parameters were then used to understand and predict the spectral response of the imbalanced MZI devices.

The fabricated MZIs with different imbalance lengths showed clear periodic transmission fringes over the measured wavelength range. The experimental results confirmed that the free spectral range (FSR) depends strongly on the imbalance length, where a larger ΔL produces a smaller FSR and more closely spaced interference fringes. The extracted effective index, group index, and FSR were compared with the simulation results. Although the overall trends showed good agreement, small deviations were observed between simulation and experiment. These differences are mainly attributed to fabrication imperfections, including waveguide width variation, etch-depth variation,

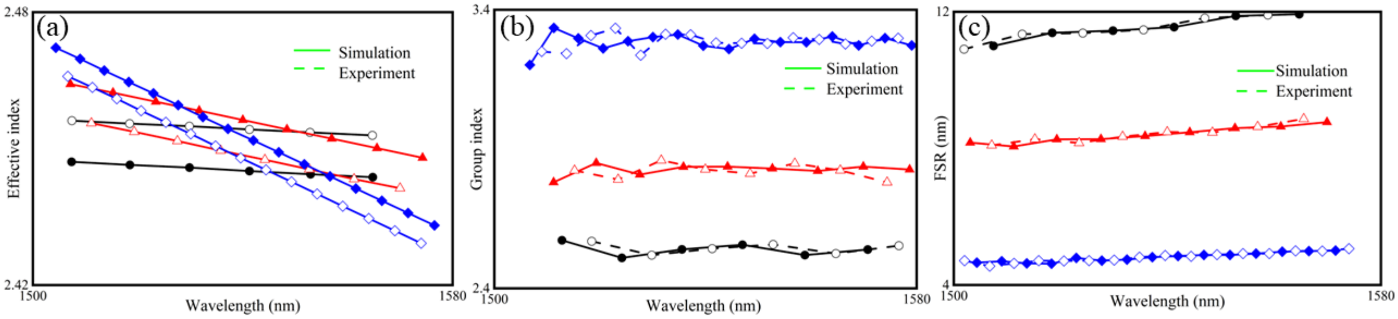

Fig. 6. Comparison of extracted optical parameters from simulation and experimental measurements for imbalanced MZIs with different imbalance lengths over the wavelength range of 1500 - 1580 nm. (a) Effective index as a function of wavelength. (b) Group index as a function of wavelength. (c) Free spectral range (FSR) as a function of wavelength. The solid curves represent simulated results, while the dashed/marker curves represent experimental results. The black, red, and blue curves correspond to MZIs with different imbalance lengths.

sidewall roughness, splitter imbalance, grating-coupler loss, and measurement noise.

Overall, the designed imbalanced MZIs successfully demonstrate the relationship between path-length difference and wavelength-dependent interference response. The results show that MZIs are useful passive test structures for extracting important waveguide parameters and evaluating fabrication tolerance in silicon photonic integrated circuits. This study provides a useful foundation for future PIC design, simulation validation, and experimental characterization of interferometric photonic devices.

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